Sample Paper 6 — Answer Key (Level-2 Difficulty Step-Up)

Total: 40 marks (30 × 1 mark + 5 × 2 marks), 60 minutes.

Section 1 — Logical Reasoning

1. **D) Shark** — Whale, Dolphin, and Bat are mammals; Shark is a fish.
2. **B) Speed** — A thermometer measures temperature; a speedometer measures speed.
3. **A) MGADQCTF** — The rule is "+2 letters" (Y wraps to A): K→M, E→G, Y→A, B→D, O→Q, A→C, R→T, D→F.
4. **C) 16** — Two patterns are interleaved: the odd-position terms go 1, 3, 5, 7 (+2 each); the even-position terms go 4, 8, 12, ? (+4 each) — so the next even term is 16.
5. **A) Aman, Chetan, Bala, Esha, Dinesh** — Bala is in the middle (position 3). Aman is at the left end (position 1). Chetan sits between them (position 2). Esha sits between Bala and Dinesh, so Esha is position 4 and Dinesh is position 5.
6. **C) Back at the starting point** — North and South cancel (8 m each way); East and West cancel (6 m each way).
7. **B) Monday** — 24 − 3 = 21 days later; 21 is exactly 3 weeks, so the day of the week repeats: Monday.
8. **D) 100** — 8 = 2×2×2, 27 = 3×3×3, 64 = 4×4×4 (each a number multiplied by itself twice — a "cube number"); 100 does not fit this pattern.
9. **D) 720** — Each term is multiplied by one more than the last multiplier used (×2, ×3, ×4, ×5, ×6): 120 × 6 = 720.
10. **C) 11th** — Adding children at the right end pushes everyone's "position from the right" up by that many: 9 + 2 = 11.

Section 2 — Mathematical Reasoning

11. **A) 63512** — 34567 + 28945 = 63512.
12. **A) 26322** — 50000 − 23678 = 26322.
13. **A) 7404** — 1234 × 6 = 7404.
14. **A) 798** — 9576 ÷ 12 = 798 exactly.
15. **C) 12** — Factors of 36: 1,2,3,4,6,9,12,18,36. Factors of 48: 1,2,3,4,6,8,12,16,24,48. HCF = 12.
16. **C) 24** — Multiples of 8: 8,16,24... Multiples of 12: 12,24... First common one = 24.
17. **A) 5/6** — As decimals: 5/6 ≈ 0.833, 7/9 ≈ 0.778, 3/4 = 0.75, 4/5 = 0.8. The greatest is 5/6.
18. **B) 4 km 250 m** — 2450 m + 1800 m = 4250 m = 4 km 250 m.

19. **C) 432 sq m** — Area = 24 × 18 = 432 sq m.
20. **B) 9 cm** — A regular pentagon has 5 equal sides: 45 ÷ 5 = 9 cm.

Section 3 — Everyday Mathematics

21. **C) ₹15,200** — 45600 ÷ 3 = 15200.
22. **B) 540** — Rate = 1440 ÷ 8 = 180 bottles/hour; in 3 hours = 180 × 3 = 540.
23. **C) 16 min** — 2400 ÷ 150 = 16 minutes.
24. **B) 4 h 20 min** — Total time 7:20 a.m. to 12:05 p.m. = 4 h 45 min; subtracting the 25-minute stop gives 4 h 20 min of actual travel.
25. **B) 850 mL** — 2500 mL + 1750 mL = 4250 mL; 4250 ÷ 5 = 850 mL per bottle.
26. **B) ₹150** — 6 × ₹28 = ₹168; ₹168 − ₹18 discount = ₹150.
27. **C) 600 L** — 1/7 × 840 = 120 L used per day; over 2 days = 240 L used; 840 − 240 = 600 L remaining.
28. **B) 56 kg 500 g** — 18500 g + 22750 g + 15250 g = 56500 g = 56 kg 500 g.
29. **C) 45** — 1080 ÷ 24 = 45 rows.
30. **B) ₹250** — Sold for 50 × ₹45 = ₹2250; profit = ₹2250 − ₹2000 = ₹250.

Section 4 — Achievers Section (2 marks each)

31. **C) 15** — Working backwards: (63 + 12) ÷ 5 = 75 ÷ 5 = 15.
32. **C) 28** — Let the first number be x. Then second = 2x, third = 2x + 10. Sum: x + 2x + (2x+10) = 150 → 5x + 10 = 150 → 5x = 140 → x = 28.
33. **C) ₹1,800** — After spending 1/3 on books, 2/3 of the money remains. After spending 1/4 of that on stationery, 3/4 of 2/3 = 1/2 of the original remains. So 1/2 of the total = ₹900 → total = ₹1,800.
34. **C) 215** — Boys = (2 × 105) + 5 = 210 + 5 = 215. Check: 105 + 215 = 320. ✓
35. **D) 50** — Each term is the sum of the previous two: 19 + 31 = 50.

Scoring guide (out of 40, Level-2-difficulty paper): 28+ here is a very strong result — it's harder than Papers 1–5 by design, so the usual 32+ Level-1 benchmark doesn't directly apply. 20–27 is solid for a first attempt at this difficulty. Below 20 — that's expected the first time through; the value here is in reviewing *why* each missed question was harder (an extra step, a bigger number, two ideas combined) rather than the raw score.